On Secure Gradient Coding with Uncoded Groupwise Keys

Xudong You[1], Kai Wan[1], Xiang Zhang[2], Wenbo Huang[1], Robert Caiming Qiu[1], and Giuseppe Caire[2]
[1]Huazhong University of Science and Technology, 430074 Wuhan, China
[2]Technische Universitat Berlin, 10587 Berlin, Germany
Emails: {xudong_you, kai_wan, eric_huang, caiming}@hust.edu.cn, {xiang.zhang, caire}@tu-berlin.de

***Abstract*—THIS PAPER IS ELIGIBLE FOR THE STUDENT PAPER AWARD. This paper considers a new secure gradient coding problem with uncoded groupwise keys, formalized as a $(\mathsf{K}, \mathsf{N}, \mathsf{N}_\mathsf{r}, \mathsf{M}, \mathsf{S})$ secure gradient coding model, where a user aims to compute the sum of the gradients from K datasets with the assistance of N distributed servers. We consider arbitrary heterogeneous data assignment, where each dataset is assigned to at least M servers. The user should recover the sum of gradients from the transmissions of any N_r servers. The security constraint guarantees that even if the user receives the transmitted messages from all servers, it cannot obtain any other information about the datasets except the sum of gradients. Compared to existing secure gradient coding works, we introduce a practical constraint on secret keys, namely uncoded groupwise keys, where the keys are mutually independent and each key is shared by precisely S servers. An achievable secure gradient coding scheme with uncoded groupwise keys is proposed, which is then proven to be optimal if $\mathsf{S} > \mathsf{M}$ and to be order optimal within a factor of 2 otherwise.**

***Index Terms*—Gradient coding, secure aggregation, uncoded groupwise keys**

I. Introduction

Distributed machine learning has become a crucial solution for training complex models on large datasets [1]–[3]. A primary challenge is the high communication cost associated with transferring large amounts of data, such as gradient vectors [3]. Another key issue is dealing with stragglers: slow servers that can severely impact computational speed [4]. Recently, coding techniques have been shown to effectively address both stragglers and adversarial server nodes, while also reducing communication overhead in distributed computations [5]–[11].

Specifically, in [12], a framework called gradient coding was introduced, enabling the master node to collect and aggregate gradient vectors with minimal communication cost, while mitigating the impact of stragglers. In this setting, homogeneous data assignment was considered, where each server is assigned the same number of datasets. Later in [13], the optimal trade-off between communication cost, computation load (i.e., the number of datasets assigned to each server), and straggler tolerance was explored. Furthermore, the authors in [14] considered arbitrary heterogeneous data assignment, where each dataset may be assigned to a different number of servers, and characterized the optimal communication cost.

The secure aggregation model for gradient coding was originally proposed in [15]. Besides the decodability constraint at the user side, an additional security constraint was added into the gradient coding problem, which imposes that the user cannot recover any other information about the datasets except the sum of gradients. For security, the servers should pre-store some keys assigned by a key distribution server. An information-theoretic lower bound on the total key size and several secure distributed coded computation schemes for specific data distribution patterns (e.g., cyclic, repetitive, and mixed) were proposed in [15], while achieving the optimal communication cost.

However, existing secure gradient coding schemes [15] rely on coded keys, which typically require a trusted key server or private links among servers, limiting their practicality in decentralized systems. Recently, uncoded groupwise keys were considered in the secure aggregation problem for federated learning (where each user holds an individual dataset) [16].[1] 'Uncoded' means that the keys are mutually independent, and 'groupwise' means that each key is shared by a subset of servers. Such keys can be generated and shared directly among servers using standard key agreement protocols [18]–[25], without requiring a trusted key server or private communication links. These properties make uncoded groupwise keys important for practical distributed learning systems.

This paper is the first which considers the secure gradient coding problem with uncoded groupwise keys. The user aims to compute the sum of gradients over K datasets with the assistance of N servers. After receiving the transmission from any N_r servers, the user should recover the sum of gradients without obtaining any other information about the datasets. By introducing a system parameter S meaning that each uncoded key is shared by exactly S servers, we aim to characterize the minimal (normalized) communication cost, for arbitrary heterogeneous data assignment where each dataset is assigned to at least M servers.

A. Main Contributions

Besides formulating the $(\mathsf{K}, \mathsf{N}, \mathsf{N}_r, \mathsf{M}, \mathsf{S})$ secure gradient coding problem with uncoded groupwise keys and arbitrary

[1]An information-theoretic secure aggregation problem using uncoded groupwise keys was formulated in [16], following the secure aggregation problem for federated learning [17]. In [16], each distributed computing node holds its own local data to compute the gradient; thus, secure aggregation against stragglers (or user dropouts) requires two-round transmissions. In contrast, in the gradient coding problem, an additional data assignment phase for servers exists, and because of the redundancy in data assignment, one-round transmission is sufficient for secure aggregation.

heterogeneous data assignment, we propose a universal secure gradient coding scheme with an explicit achievable communication cost. Our scheme jointly exploits data redundancy and key redundancy to achieve the minimum communication cost required for secure gradient aggregation. Compared to the communication cost of the optimal non-secure gradient coding scheme, the proposed scheme achieves the same communication cost when $\mathsf{S} > \mathsf{M}$, and achieves a communication cost within a multiplicative gap of factor 2 when $\mathsf{S} \leq \mathsf{M}$.

Paper Organization. This paper is organized as follows. Section II introduces the secure gradient coding model with uncoded groupwise keys. Section III presents the main results. Section IV describes the proposed general coding scheme. Finally, section V concludes this paper.

Notation. Calligraphic symbols denote sets, bold symbols denote vectors and matrices, and sans-serif symbols denote system parameters. We use $|\cdot|$ to represent the cardinality of a set or the length of a vector; $[a:b] := \{a, a+1, \ldots, b\}$ and $[n] := [1:n]$; $\mathbf{1}_n$ and $\mathbf{0}_n$ represent the vertical n-dimensional vector whose elements are all 1 and all 0, respectively; $\mathbf{A}^\mathsf{T}$ and $\mathbf{A}^{-1}$ represent the transpose and the inverse of matrix $\mathbf{A}$, respectively; the matrix $[\mathbf{A};\mathbf{B}]$ is expressed in a format similar to Matlab, equivalent to $\begin{bmatrix}\mathbf{A}\\ \mathbf{B}\end{bmatrix}$;$\mathbf{A}(\mathcal{N},\cdot)$ and $\mathbf{A}(\cdot,\mathcal{O})$ represent the sub-matrices of matrix $\mathbf{A}$ whose rows and columns in the set $\mathcal{N}$, respectively; $\mathbf{A}(\mathcal{N},\mathcal{O})$ represents the sub-matrix of matrix $\mathbf{A}$ whose rows are in the set $\mathcal{N}$ and columns are in the set $\mathcal{O}$; $\mathbf{I}_n$ represents the identity matrix of dimension $n \times n$; $\mathbf{0}_{m\times n}$ represents all-zero matrix of dimension $m \times n$; $\mathbf{A}_{m\times n}$ explicitly indicates that the matrix $\mathbf{A}$ is of dimension $m \times n$; let $\binom{x}{y} = 0$ if $x < 0$ or $y < 0$ or $x < y$; let $\binom{\mathcal{X}}{y} = \{\mathcal{S} \subseteq \mathcal{X} : |\mathcal{S}| = y\}$, where $|\mathcal{X}| \geq y > 0$; For a set $\mathcal{S}$, we denote the i^{th} smallest element by $\mathcal{S}(i)$; $\mathbb{F}_\mathsf{q}$ represents a finite field with order q. In the rest of the paper, entropies will be in base q, where q represents the field size.

II. System Model

Given a training dataset $D = \{(x_i, y_i)\}_{i=1}^d$ and let $\boldsymbol{\beta}$ be the model parameters. Given a loss function $\mathcal{L}(\cdot)$, the goal of training is to minimize:

$$\mathcal{L}(D;\boldsymbol{\beta}) = \sum_{i=1}^{d} \mathcal{L}(x_i, y_i; \boldsymbol{\beta}). \tag{1}$$

Gradient-based methods update the model parameters using the full gradient $\mathbf{g} = \nabla_{\boldsymbol{\beta}} \mathcal{L}(D;\boldsymbol{\beta})$, which is repeatedly evaluated during training. To enable distributed computation, the dataset D is partitioned into K disjoint subsets $\{D_1, \ldots, D_\mathsf{K}\}$. Then the main task is to compute the sum of gradients $\mathbf{g} = \sum_{k=1}^{\mathsf{K}} \mathbf{g}_k$, where $g_k \in \mathbb{F}_q^{\mathsf{L}}$ is the gradient over D_k for $k \in [\mathsf{K}]$. Assume that the gradients are mutually independent.

Then we formulate the $(\mathsf{K}, \mathsf{N}, \mathsf{N_r}, \mathsf{M}, \mathsf{S})$ secure gradient coding problem with uncoded groupwise keys and arbitrary heterogeneous data assignment. A user aims to compute the sum of gradients over K datasets $\{D_1, \ldots, D_\mathsf{K}\}$ with the assistance of any $\mathsf{N_r}$ servers selected from a total of N servers, while learning no additional information about the individual gradients. For each subset $\mathcal{V} \subseteq [\mathsf{N}]$ with $|\mathcal{V}| = \mathsf{S}$, a random key $Q_\mathcal{V}$ is generated and shared exclusively among the servers in $\mathcal{V}$ using key agreement protocols. These keys are referred to as uncoded groupwise keys. All keys are mutually independent and are also independent of the gradients, i.e.,

$$\begin{aligned} &H\left(\left(Q_\mathcal{V} : \mathcal{V} \in \binom{[\mathsf{N}]}{\mathsf{S}}\right), (g_1, \ldots, g_\mathsf{K})\right) \\ &= \sum_{\mathcal{V}\in\binom{[\mathsf{N}]}{\mathsf{S}}} H(Q_\mathcal{V}) + \sum_{k\in[\mathsf{K}]} H(g_k). \end{aligned} \tag{2}$$

A secure gradient coding scheme consists of three phases: data assignment, encoding, and decoding.

Data Assignment: Each dataset D_k is assigned to a subset of servers $\mathcal{D}_k \subseteq [\mathsf{N}]$ with $|\mathcal{D}_k| \geq \mathsf{M}$, where M denotes the minimum replication factor. For each server $n \in [\mathsf{N}]$, $\mathcal{Z}_n \subseteq [\mathsf{K}]$ denotes the set of datasets assigned to it. Assume that $\mathcal{Z}_1, \ldots, \mathcal{Z}_\mathsf{N}$ are pre-fixed; that is, we consider arbitrary heterogeneous data assignment.

Encoding: Each server $n \in [\mathsf{N}]$ computes the gradients g_k locally for all $k \in \mathcal{Z}_n$. Based on these gradients and the assigned keys, server n generates a coded message X_n, which is then transmitted to the user, i.e.,

$$H\left(X_n | \left(g_k : k \in \mathcal{Z}_n\right), \left(Q_\mathcal{V} : n \in \mathcal{V}\right)\right) = 0, \forall n \in [\mathsf{N}]. \tag{3}$$

Decoding: The user collects coded messages from any subset of servers $\mathcal{U} \subseteq [\mathsf{N}]$ with $|\mathcal{U}| = \mathsf{N_r}$. Based on the received messages $\mathcal{X} = \{X_n : n \in \mathcal{U}\}$, the user must be able to recover the sum of gradients:

$$H\left(\sum_{k\in[\mathsf{K}]} g_k | \left(X_n : n \in \mathcal{U}\right)\right) = 0, \forall \mathcal{U} \in \binom{[\mathsf{N}]}{\mathsf{N_r}}. \tag{4}$$

To enable cancellation of all keys through linear operations, each key must appear in at least two received messages. Otherwise, a key that appears only once cannot be eliminated. This requirement leads to the following feasibility condition:

$$\mathsf{S} \geq \mathsf{N} - \mathsf{N_r} + 2. \tag{5}$$

In addition to decodability, the scheme must satisfy an information-theoretic security requirement. Specifically, even if the user received transmissions from all servers, it should not be able to obtain any information about the individual gradients beyond their sum. This requirement is formalized as:

$$I\left(X_1, \ldots, X_\mathsf{N}; g_1, \ldots, g_\mathsf{K} | \sum_{k\in[\mathsf{K}]} g_k\right) = 0. \tag{6}$$

The communication cost is defined as the maximum normalized message size transmitted by any server:

$$\mathsf{R} \triangleq \max_{n\in[\mathsf{N}]} \frac{|X_n|}{\mathsf{L}}. \tag{7}$$

The objective is to characterize the optimal communication cost for arbitrary heterogeneous data assignment R^*.

The optimal communication cost of non-secure gradient coding under linear coding was characterized in [14].

Lemma 1 ([14]): For the $(\mathsf{K},\mathsf{N},\mathsf{N_r},\mathsf{M})$ non-secure gradient coding model with arbitrary data assignment, the optimal communication cost under linear coding is equal to

$$\mathsf{R}^*_{\mathrm{n}} = \frac{1}{\mathsf{N_r}-\mathsf{N}+\mathsf{M}}. \tag{8}$$

Clearly, we have $\mathsf{R}^* \le \mathsf{R}^*_{\mathrm{n}}$. Note that it was shown in [15] that if there is no constraint on the shared keys at the users, adding security guaranty on gradient coding does not induce additional communication cost. However, this paper considers the constraint of uncoded groupwise keys.

III. Main Results

For the $(\mathsf{K},\mathsf{N},\mathsf{N_r},\mathsf{M},\mathsf{S})$ secure gradient coding problem with uncoded groupwise keys and arbitrary heterogeneous data assignment, this paper proposes a universal secure gradient coding scheme as follows, whose proof could be found in Section IV.

Theorem 1 (Achievable Communication Cost): For the $(\mathsf{K},\mathsf{N},\mathsf{N_r},\mathsf{M},\mathsf{S})$ secure gradient coding problem with uncoded groupwise keys and arbitrary heterogeneous data assignment, the following communication cost is achievable:

$$\mathsf{R} = \frac{\binom{\mathsf{N}}{\mathsf{S}}-\binom{\mathsf{M}}{\mathsf{S}}}{\left(\binom{\mathsf{N}}{\mathsf{S}}-\binom{\mathsf{M}}{\mathsf{S}}\right)\mathsf{N_r}-\binom{\mathsf{N}}{\mathsf{S}}(\mathsf{N}-\mathsf{M})}. \tag{9}$$

The communication cost is determined by how many groupwise keys cannot be canceled using data redundancy. When fewer servers share a key, additional communication is required to eliminate it. Larger key groups increase overlap and reduce this overhead.

We next provide the (order) optimality results of the proposed scheme, whose proof could be found in Appendix A.

Theorem 2 ((Order) optimality)): When $\mathsf{S} > \mathsf{M}$, the achievable communication cost coincides with the optimal communication cost of non-secure gradient coding under linear coding,

$$\mathsf{R}^* = \mathsf{R}^*_{\mathrm{n}} = \frac{1}{\mathsf{N_r}-\mathsf{N}+\mathsf{M}}; \tag{10}$$

when $\mathsf{N}-\mathsf{N_r}+2 \le \mathsf{S} \le \mathsf{M}$, the achievable communication cost is order-optimal within a factor of 2,

$$\mathsf{R} \le \mathsf{R}^*_{\mathrm{n}} \le 2\mathsf{R}^*. \tag{11}$$

Note that when $\mathsf{S} \le \mathrm{M}$, limited key sharing prevents full cancellation through data redundancy, leading to extra communication. However, this overhead is bounded, and the communication cost remains within a constant factor of the optimum. It then reveals a sharp transition at $\mathsf{S}=\mathsf{M}$: above this threshold, security is essentially free, while below it, security incurs an order-optimal communication overhead.

Numerical evaluation: We then compare our communication cost with the optimal cost of non-secure gradient coding in [14]. Fig. 1 shows that the communication cost decreases as M increases. When $\mathsf{S} > \mathsf{M}$, our scheme matches the optimal cost, confirming that sufficient key overlap eliminates extra overhead. Fig. 2 illustrates the effect of S. With small S, the cost is higher due to limited key sharing, but it quickly drops as S grows. When $\mathsf{S} \ge \mathsf{M}$, the cost achieves the optimum. Overall, the numerical results demonstrate that our scheme remains efficient across parameter settings and achieves near-optimal performance.

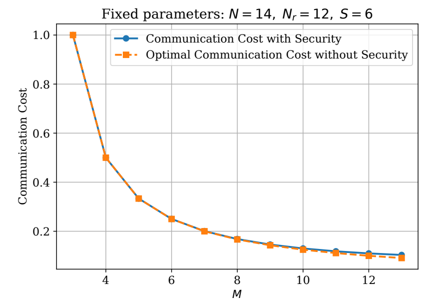

Fig. 1. $\mathsf{N}=14, \mathsf{N_r}=12, \mathsf{S}=6$, with varying $\mathsf{M} \in [3:13]$.

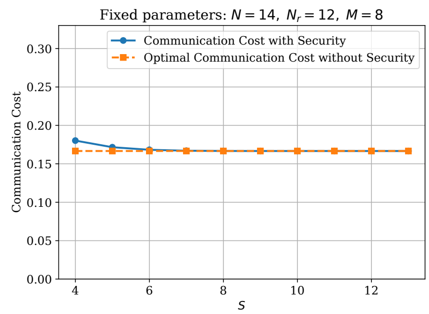

Fig. 2. $\mathsf{N}=14, \mathsf{N_r}=12, \mathsf{M}=8$, with varying $\mathsf{S} \in [4:13]$.

IV. Proof of Theorem 1: Achievable scheme

A. Example: $(\mathsf{K},\mathsf{N},\mathsf{N_r},\mathsf{M},\mathsf{S}) = (3,3,3,2,2)$

We first illustrate the proposed scheme through an example, where $(\mathsf{K},\mathsf{N},\mathsf{N_r},\mathsf{M},\mathsf{S}) = (3,3,3,2,2)$.

Data assignment: Each dataset D_i is assigned to two distinct servers. Specifically, in this example, $\mathcal{D}_1=\{2,3\}, \mathcal{D}_2=\{1,2\}, \mathcal{D}_3=\{1,2\}$. For each $\mathcal{V} \in \binom{[3]}{2}$, the servers in $\mathcal{V}$ share a key $Q_{\mathcal{V}}$ with $\frac{\mathsf{L}}{3}$ uniformly i.i.d. symbols on $\mathbb{F}_q$. In this example, for the ease of illustration, we assume that q is a large prime number, which is not required in our general scheme. The assignment of datasets and keys is illustrated in Fig. 3.

Encoding: To achieve our communication cost $\mathsf{R}=2/3$ in Theorem 1, each gradient g_k is partitioned into 3 non-overlapping and equal-length pieces $g_k = (g_{k,1}, g_{k,2}, g_{k,3})$, each piece containing $\mathsf{L}/3$ symbols. Each server then transmits 2 independent linear combinations of pieces and keys to the user. After the user receives 6 transmissions from three servers,

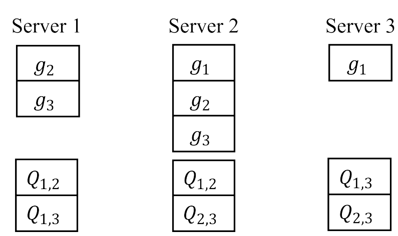

Fig. 3. Data and key assignment for $(\mathsf{K},\mathsf{N},\mathsf{N_r},\mathsf{M},\mathsf{S}) = (3,3,3,2,2)$.

the user can recover $\mathbf{FW}$. More precisely, $\mathbf{F}$ is a matrix with dimension 6×12,

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}_1 & \mathbf{0}_{3\times 3} \\ \mathbf{F}_2 & \mathbf{F}_3 \end{bmatrix}, \tag{12}$$

where $\mathbf{F}_1$ specifies the computation demand

$$\mathbf{F}_1 = \begin{bmatrix} 111 & 000 & 000 \\ 000 & 111 & 000 \\ 000 & 000 & 111 \end{bmatrix}_{3\times 9}, \tag{13}$$

and we set $\mathbf{F}_3 = \mathbf{I}_3$ while $\mathbf{F}_2$ will be designed later. $\mathbf{W}$ is a matrix composed of gradient pieces and keys given by

$$\mathbf{W} = [g_{1,1}; g_{2,1}; g_{3,1}; g_{1,2}; \dots; g_{3,3}; Q_{1,2}; Q_{1,3}; Q_{2,3}]. \tag{14}$$

Each server $n \in [3]$ transmits a coded message $X_n = \mathbf{C}_n\mathbf{FW}$, where $\mathbf{C}_n$ is an encoding matrix with dimension 2×6 to be determined.

Decoding: After receiving transmissions $\mathcal{X} = \{X_n : n \in [3]\}$ from servers in $\mathcal{U} = \{1, 2, 3\}$, the user can decode the received messages as:

$$\mathbf{C}_{\mathcal{U}}^{-1}\mathcal{X} = \begin{bmatrix} \mathbf{F}_1 & \mathbf{0} \\ \mathbf{F}_2 & \mathbf{I}_3 \end{bmatrix}\mathbf{W}, \tag{15}$$

where $\mathbf{C}_{\mathcal{U}} = [\mathbf{C}_1; \mathbf{C}_2; \mathbf{C}_3]$ and the sum of gradients $g_1 + g_2 + g_3$ is recovered from the first three rows of $\mathbf{FW}$.

To guarantee the successful transmission at each server, the correct recovery of $\mathbf{FW}$, and the information-theoretic security of the individual gradients, we impose some design constraints on $(\mathbf{C}_n : n \in [3])$ and $\mathbf{F}_2$, which we describe next.

Encodability constraint. To ensure that each server $n \in [\mathsf{N}]$ only transmits the gradients and keys assigned to it, we have

$$\mathbf{C}_n\mathbf{F}(\cdot, \overline{\mathcal{C}_n}) = \mathbf{0}_{2\times|\overline{\mathcal{C}_n}|}, \tag{16}$$

where r denotes the number of linear combinations of gradient pieces and keys transmitted by each server (in this example $r = 2$), and $\overline{\mathcal{C}_n}$ denotes the set of columns in $\mathbf{F}$ corresponding to gradients (or gradient pieces) and keys not assigned to server n. This condition guarantees that in the message X_n sent by server $n \in [\mathsf{N}]$, the coefficients corresponding to the unassigned gradients (or gradient pieces) and keys are 0.

Decodability constraint. The user should be able to recover $\mathbf{FW}$ from the transmissions of any $\mathsf{N_r}$ servers. Accordingly, for any $\mathcal{U} \in \binom{[\mathsf{N}]}{\mathsf{N_r}}$, the matrix

$$\mathbf{C}_{\mathcal{U}} = \begin{bmatrix} \mathbf{C}_{\mathcal{U}(1)} \\ \vdots \\ \mathbf{C}_{\mathcal{U}(\mathsf{N_r})} \end{bmatrix} \tag{17}$$

should be full rank.

Now it remains to design $\mathbf{C}$ and $\mathbf{F}_2$ satisfying the encodability and decodability constraints. In the following, we will construct $\mathbf{C}$ to satisfy the encodability constraint of key part and decodability constraint, and satisfy the encodability constraint of gradient part by the design of $\mathbf{F}_2$.

1) Design Matrix $\mathbf{C}$: For server 1, who has the keys $Q_{1,2}$ and $Q_{1,3}$ without knowing $Q_{2,3}$, its transmitted coefficients associated with unassigned keys $Q_{2,3}$ should be 0, which imposes the following constraint:

$$\mathbf{C}_1\mathbf{F}(\cdot, \{12\}) = \mathbf{C}_1 \left[0, 0, 0, 0, 0, 1\right]^{\mathsf{T}} = \mathbf{0}_{2\times 1}. \tag{18}$$

This constraint can be satisfied by generating $\mathbf{C}_1$ from the left nullspace of $\mathbf{F}(\cdot, \{12\})$. Applying the same principle to the remaining servers, we complete the construction of $\mathbf{C}$, which is full rank and thus satisfies the decodability constraint.

$$\mathbf{C} = \begin{bmatrix} * & * & * & * & * & 0 \\ * & * & * & * & * & 0 \\ * & * & * & * & 0 & * \\ * & * & * & * & 0 & * \\ * & * & * & 0 & * & * \\ * & * & * & 0 & * & * \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 3 & 4 & 0 & 1 & 0 \\ 3 & 2 & 5 & 1 & 0 & 0 \\ 8 & 5 & 7 & 0 & 0 & 1 \\ 9 & 7 & 2 & 0 & 1 & 0 \\ 7 & 4 & 1 & 0 & 0 & 1 \end{bmatrix},$$

where '*' represents an i.i.d. uniform element on $\mathbb{F}_q$.

2) Design Matrix $\mathbf{F}_2$: For gradient g_1, which is stored only at servers 2 and 3, server 1 must not transmit any piece of g_1. This requires

$$\mathbf{C}_1\mathbf{F}(\cdot, \mathcal{G}_1) = \begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 3 & 4 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} = \mathbf{0}_{2\times 3}, \tag{19}$$

where $\mathcal{G}_1 = \{1+3(i-1) : i \in [3]\}$ denotes the set of columns in $\mathbf{F}$ corresponding to gradient g_1. Since $\mathbf{C}_1(\cdot, \{4, 5, 6\})$ is left full rank, this system is solvable. By applying the same procedure to the other gradients, we obtain the complete construction of $\mathbf{F}_2$.

At this point, both $\mathbf{C}$ and $\mathbf{F}_2$ have been fully specified and satisfy the constraints.

Security: We provide an intuitive explanation of this security constraint. The overall information obtained by the user, $\mathbf{FW}$, can be horizontally partitioned into two blocks. The first block $[\mathbf{F}_1, \mathbf{0}]\mathbf{W}$ is the desired sum of gradients $\mathbf{F}_1[g_{1,1}; \dots; g_{3,3}]$. Hence, the security is determined by the second block

$$[\mathbf{F}_2, \mathbf{F}_3]\mathbf{W} = \begin{bmatrix} * & \cdots & * & 1 & 0 & 0 \\ * & \cdots & * & 0 & 1 & 0 \\ * & \cdots & * & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} g_{1,1} \\ \vdots \\ g_{3,3} \\ Q_{1,2} \\ Q_{1,3} \\ Q_{2,3} \end{bmatrix}, \tag{20}$$

where $\mathbf{F}_2$ multiplies individual gradients and must be protected, while $\mathbf{F}_3$ multiplies the keys. If $\mathbf{F}_3$ is full rank, no linear transformation can eliminate any of its rows. As a result, each row of $\mathbf{F}_2$ is always masked by at least one independent key. As $\mathbf{F}_3$ is designed as an identity matrix, the security constraint is satisfied.

B. General scheme

We are now ready to describe our general scheme for the $(\mathsf{K}, \mathsf{N}, \mathsf{N_r}, \mathsf{M}, \mathsf{S})$ secure gradient coding model. It has been proved in [26] that, by assuming that L is large enough, we can directly set that q is large enough. Denote the communication cost in (9) as $\mathsf{R} = r/n$ for simplicity.

Data assignment: The datasets are assigned to servers in an arbitrary way. Each dataset D_k is replicated at least M times among the servers in the set $\mathcal{D}_k$. The keys are generated in an uncoded groupwise manner. Each group $\mathcal{V} \in \binom{[\mathsf{N}]}{\mathsf{S}}$ shares a unique key $Q_{\mathcal{V}}$. To simplify notation, we replace the key notation from set $Q_{\mathcal{V}}$ to index Q_i, where $\mathcal{V}$ is the i-th set in $\binom{[\mathsf{N}]}{\mathsf{S}}$. Each key has $\mathsf{L}(\mathsf{N}-\mathsf{M})/n$ i.i.d. uniform symbols in $\mathbb{F}_q$.

Encoding: To achieve our communication cost $\mathsf{R} = r/n$, each gradient g_k is partitioned into n non-overlapping and equal-length pieces $g_k = (g_{k,1}, \ldots, g_{k,n})$. each Q_i is further partitioned into $\mathsf{N}-\mathsf{M}$ non-overlapping and equal-length pieces $Q_i = (Q_{i,1}, \ldots, Q_{i,\mathsf{N}-\mathsf{M}})$. So each gradient piece and each key piece both contain L/n i.i.d. uniform symbols in $\mathbb{F}_q$. Each server then transmits r independent linear combinations of gradient and key pieces to the user. After the user receives $r\mathsf{N}_\mathsf{r}$ transmissions from any N_r servers, the user can recover $\mathbf{FW}$. The demand matrix $\mathbf{F}$ is defined as:

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}_1 & \mathbf{0}_{n\times(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}} \\ \mathbf{F}_2 & \mathbf{F}_3 \end{bmatrix}, \quad (21)$$

where $\mathbf{F}_1$ specifies the computation demand

$$\begin{bmatrix} \mathbf{1}_\mathsf{K} & \cdots & \mathbf{0}_\mathsf{K} \\ \vdots & \ddots & \vdots \\ \mathbf{0}_\mathsf{K} & \cdots & \mathbf{1}_\mathsf{K} \end{bmatrix}_{n\times n\mathsf{K}}, \quad (22)$$

and we set $\mathbf{F}_3 = \mathbf{I}_{(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}}$ while $\mathbf{F}_2$ will be designed later.

The message matrix $\mathbf{W}$ is given by

$$\mathbf{W} = [g_{1,1}; g_{2,1}; ...; g_{\mathsf{K},n}; Q_{1,1}; Q_{2,1}; ...; Q_{\binom{\mathsf{N}}{\mathsf{S}},\mathsf{N}-\mathsf{M}}]. \quad (23)$$

Each server $n \in [\mathsf{N}]$ transmits a coded message

$$X_n = \mathbf{C}_n\mathbf{FW}, \quad (24)$$

where $\mathbf{C}_n$ is an encoding matrix with dimension $r \times r\mathsf{N}_\mathsf{r}$.

Decoding: After receiving $\mathcal{X} = \{X_n : n \in \mathcal{U}\}$ from servers in $\mathcal{U} \in \binom{[\mathsf{N}]}{\mathsf{N}_\mathsf{r}}$, the user can decode the received messages as:

$$\mathbf{C}_{\mathcal{U}}^{-1}\mathcal{X} = \begin{bmatrix} \mathbf{F}_1 & \mathbf{0} \\ \mathbf{F}_2 & \mathbf{I} \end{bmatrix}\mathbf{W}, \quad (25)$$

from which the sum of gradients $g_1 + \ldots + g_\mathsf{K}$ is recovered from the first n rows of $\mathbf{FW}$.

To ensure the correctness of the proposed scheme, the encodability and decodability constraints introduced in Section IV-A should be satisfied, by designing $\mathbf{C}$ and $\mathbf{F}_2$.

For each server $n \in [\mathsf{N}]$, let $\mathcal{S}_n \subseteq \binom{\mathsf{N}}{\mathsf{S}}$ denote the set of indices of keys available to server n, and let $\overline{\mathcal{S}_n}$ denote its complement. In the transmitted message X_n, the coefficients corresponding to unassigned keys $\overline{\mathcal{S}_n}$ are zero; thus

$$\mathbf{C}_n\mathbf{F}(\cdot, \overline{\mathcal{H}_n}) = \mathbf{0}, \quad (26)$$

where $\overline{\mathcal{H}_n} = \{n\mathsf{K} + i + (j-1)\binom{\mathsf{N}}{\mathsf{S}} : i \in \overline{\mathcal{S}_n}, j \in [\mathsf{N}-\mathsf{M}]\}$ indexes the columns of $\mathbf{F}$ corresponding to keys unavailable to server n. We construct $\mathbf{C}_n$ by selecting its rows from the left nullspace of $\mathbf{F}(\cdot, \overline{\mathcal{H}_n})$. Applying this procedure to all servers yields the complete encoding matrix $\mathbf{C}$, which is full rank and thus satisfies the decodability constraint.[2]

After determining $\mathbf{C}$, we determine $\mathbf{F}_2$. For each gradient g_k, which is known by the servers in $\mathcal{D}_k$, the servers in $\overline{\mathcal{D}_k}$ cannot transmit the corresponding gradient pieces; thus

$$\mathbf{C}(\overline{\mathcal{D}_k}, \cdot)\mathbf{F}(\cdot, \mathcal{G}_k) = \mathbf{0}, \quad (27)$$

where $\mathcal{G}_k = \{k + (i-1)\mathsf{K} : i \in [n]\}$ indexes the columns of $\mathbf{F}$ corresponding to g_k. These constraints reduce to affine linear systems in $\mathbf{F}_2$. Since $\mathbf{C}(\overline{\mathcal{D}_k}, \mathcal{O})$ is left full rank,[3] where $\mathcal{O}$ is the set of columns in $\mathbf{C}$ multiplied by $\mathbf{F}_2$, these constraints are solvable. Solving these systems for all $k \in [\mathsf{K}]$ completes the construction of $\mathbf{F}$.

At this point, both $\mathbf{C}$ and $\mathbf{F}_2$ have been fully specified and satisfy both constraints.

Proof of security. We provide an intuitive proof on the security in the following, where the formal information theoretic proof is provided in Appendix D. The overall information obtained by the user, $\mathbf{FW}$, can be horizontally partitioned into two blocks as in the above example. The first block $[\mathbf{F}_1, \mathbf{0}]\mathbf{W}$ is the computation task. The security is mainly concentrated on second block $[\mathbf{F}_2, \mathbf{F}_3]\mathbf{W}$, where $\mathbf{F}_2$ contains the information of individual gradients and must be protected. Since $\mathbf{F}_3$ is an identity matrix, no linear transformation can eliminate any of its rows. Hence, each row of $\mathbf{F}_2$ is always protected by at least one independent key, which satisfies the security constraint.

Communication cost. We derive the communication cost from the encodability constraint of the gradient part, which requires $\mathbf{C}(\overline{\mathcal{D}_k}, \mathcal{O})$ to be full-rank. Assume $\mathsf{R} = \frac{r}{n}$ and each key is partitioned into α pieces. The dimension of the available subspace for $\mathbf{C}(\overline{\mathcal{D}_k}, \mathcal{O})$ is $\alpha(\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{M}}{\mathsf{S}})$,[4] which must be no smaller than the number of rows, yielding the constraint $\alpha\left(\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{M}}{\mathsf{S}}\right) \geq (\mathsf{N}-\mathsf{M})r$. The coding matrix $\mathbf{C}$ has dimension $r\mathsf{N} \times r\mathsf{N}_\mathsf{r}$, and the demand matrix $\mathbf{F}$ has dimension $(n + \alpha\binom{\mathsf{N}}{\mathsf{S}}) \times (n\mathsf{K} + \alpha\binom{\mathsf{N}}{\mathsf{S}})$. For the product $\mathbf{CF}$ to be well defined, we have $r\mathsf{N}_\mathsf{r} = n + \alpha\binom{\mathsf{N}}{\mathsf{S}}$. Finally, eliminating α, we obtain the communication cost in Theorem 1.

V. Conclusion

In this paper, we proposed a novel secure gradient coding model with uncoded groupwise keys. Under arbitrary data assignment, we present a general communication cost together with an achievable scheme. The communication cost is shown to be optimal when $S > M$ and order-optimal within a factor of 2 of the optimum when $S \leq M$, where S and M denote the key group size and minimum data replication factor, respectively. Our results identify a sharp transition in which security incurs no additional communication overhead once the key group size exceeds the data replication factor, providing practical design guidelines for secure and communication-efficient distributed learning systems.

[2] $\mathbf{C}_n$ is generated from the nullspace of $\mathbf{F}(\cdot, \overline{\mathcal{H}_n})$, whose available dimension is proportional to the keys available to server n. For any servers in $\mathcal{U} \in \binom{[\mathsf{N}]}{\mathsf{N}_\mathsf{r}}$, all keys are available, ensuring $\mathbf{C}_{\mathcal{U}}$ contains sufficient independent vectors. Consequently, $\mathbf{C}_{\mathcal{U}}$ is full rank with high probability. The detailed decodability proof is provided in Appendix C.

[3] This property is ensured by our communication cost design, which is derived from the requirement that $\mathbf{C}(\overline{\mathcal{D}_k}, \mathcal{O})$ is left full rank. The detailed proof is in Appendix B.

[4] The detailed derivation is provided in Appendix E.

References

[1] J. Dean, G. Corrado, R. Monga, K. Chen, M. Devin, M. Mao, M. Ranzato, A. Senior, P. Tucker, K. Yang *et al.*, "Large scale distributed deep networks," *Advances in neural information processing systems*, vol. 25, 2012.

[2] A. Ahmed, N. Shervashidze, S. Narayanamurthy, V. Josifovski, and A. J. Smola, "Distributed large-scale natural graph factorization," in *Proceedings of the 22nd international conference on World Wide Web*, 2013, pp. 37–48.

[3] M. Li, D. G. Andersen, A. J. Smola, and K. Yu, "Communication efficient distributed machine learning with the parameter server," *Advances in Neural Information Processing Systems*, vol. 27, 2014.

[4] J. Dean and L. A. Barroso, "The tail at scale," *Communications of the ACM*, vol. 56, no. 2, pp. 74–80, 2013.

[5] S. Li, M. A. Maddah-Ali, and A. S. Avestimehr, "Coded mapreduce," in *2015 53rd Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, 2015, pp. 964–971.

[6] K. Lee, M. Lam, R. Pedarsani, D. Papailiopoulos, and K. Ramchandran, "Speeding up distributed machine learning using codes," *IEEE Trans. Inf. Theory*, vol. 64, no. 3, Mar. 2018.

[7] Q. Yu, S. Li, N. Raviv, S. M. M. Kalan, M. Soltanolkotabi, and S. A. Avestimehr, "Lagrange coded computing: Optimal design for resiliency, security, and privacy," in *The 22nd International Conference on Artificial Intelligence and Statistics*. PMLR, 2019, pp. 1215–1225.

[8] Q. Yu, M. Maddah-Ali, and S. Avestimehr, "Polynomial codes: an optimal design for high-dimensional coded matrix multiplication," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[9] Q. Yu, M. A. Maddah-Ali, and A. S. Avestimehr, "Straggler mitigation in distributed matrix multiplication: Fundamental limits and optimal coding," *IEEE Transactions on Information Theory*, vol. 66, no. 3, pp. 1920–1933, 2020.

[10] S. Dutta, M. Fahim, F. Haddadpour, H. Jeong, V. Cadambe, and P. Grover, "On the optimal recovery threshold of coded matrix multiplication," *IEEE Transactions on Information Theory*, vol. 66, no. 1, pp. 278–301, 2019.

[11] S. Dutta, V. Cadambe, and P. Grover, "Short-dot: Computing large linear transforms distributedly using coded short dot products," *Advances In Neural Information Processing Systems*, vol. 29, 2016.

[12] R. Tandon, Q. Lei, A. G. Dimakis, and N. Karampatziakis, "Gradient coding: Avoiding stragglers in distributed learning," in *International Conference on Machine Learning*. PMLR, 2017, pp. 3368–3376.

[13] M. Ye and E. Abbe, "Communication-computation efficient gradient coding," in *International Conference on Machine Learning*. PMLR, 2018, pp. 5610–5619.

[14] T. Jahani-Nezhad and M. A. Maddah-Ali, "Optimal communication-computation trade-off in heterogeneous gradient coding," *IEEE Journal on Selected Areas in Information Theory*, vol. 2, no. 3, pp. 1002–1011, 2021.

[15] K. Wan, H. Sun, M. Ji, and G. Caire, "On secure distributed linearly separable computation," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 3, pp. 912–926, 2022.

[16] K. Wan, X. Yao, H. Sun, M. Ji, and G. Caire, "On the information theoretic secure aggregation with uncoded groupwise keys," *IEEE Transactions on Information Theory*, vol. 70, no. 9, pp. 6596–6619, 2024.

[17] K. Bonawitz, V. Ivanov, B. Kreuter, A. Marcedone, H. B. McMahan, S. Patel, D. Ramage, A. Segal, and K. Seth, "Practical secure aggregation for privacy-preserving machine learning," in *proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 2017, pp. 1175–1191.

[18] W. Diffie and M. E. Hellman, "New directions in cryptography," in *Democratizing Cryptography: The Work of Whitfield Diffie and Martin Hellman*, 2022, pp. 365–390.

[19] U. M. Maurer, "Secret key agreement by public discussion from common information," *IEEE transactions on information theory*, vol. 39, no. 3, pp. 733–742, 1993.

[20] R. Ahlswede and I. Csiszár, "Common randomness in information theory and cryptography. i. secret sharing," *IEEE Transactions on Information Theory*, vol. 39, no. 4, pp. 1121–1132, 1993.

[21] I. Csiszár and P. Narayan, "Secrecy capacities for multiple terminals," *IEEE Transactions on Information Theory*, vol. 50, no. 12, pp. 3047–3061, 2004.

[22] A. A. Gohari and V. Anantharam, "Information-theoretic key agreement of multiple terminals—part i," *IEEE Transactions on Information Theory*, vol. 56, no. 8, pp. 3973–3996, 2010.

[23] H. Sun, "Secure groupcast with shared keys," *IEEE Transactions on Information Theory*, vol. 68, no. 7, pp. 4681–4699, 2022.

[24] ——, "Compound secure groupcast: Key assignment for selected broadcasting," *IEEE Journal on Selected Areas in Information Theory*, vol. 3, no. 2, pp. 379–389, 2022.

[25] Z. Li, X. Zhang, Y. Zhao, H. Chen, J. Fan, and G. Caire, "The capacity of collusion-resilient decentralized secure aggregation with groupwise keys," *arXiv preprint arXiv:2511.14444*, 2025.

[26] Y. Zhao and H. Sun, "Information theoretic secure aggregation with user dropouts," *IEEE Transactions on Information Theory*, vol. 68, no. 11, pp. 7471–7484, 2022.

Appendix A
Order-Optimal Proof

We give the order-optimal proof of our scheme.

When $\mathsf{S} > \mathsf{M}$, we have $\binom{\mathsf{M}}{\mathsf{S}} = 0$. Substituting this into (9) shows that the achievable communication cost reduces exactly to the optimal expression in Lemma 1.

Let us then consider the case where $\mathrm{N} - \mathrm{N_r} + 2 \leq \mathsf{S} \leq \mathsf{M}$. We compare the achievable communication cost with the non-secure optimum by examining their ratio:

$$\frac{\mathsf{R}}{\mathsf{R}^*} = \frac{\mathsf{N_r} - (\mathsf{N} - \mathsf{M})}{\mathsf{N_r} - \beta(\mathsf{N} - \mathsf{M})}, \tag{28}$$

where $\beta = \frac{\binom{\mathsf{N}}{\mathsf{S}}}{\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{M}}{\mathsf{S}}}$.

From the above expression, $\frac{\mathsf{R}}{\mathsf{R}^*}$ is maximized when β is maximized. We therefore analyze β, which can be rewritten as:

$$\beta = \frac{1}{1 - \frac{\mathsf{M}(\mathsf{M}-1)\ldots(\mathsf{M}-\mathsf{S}+1)}{\mathsf{N}(\mathsf{N}-1)\ldots(\mathsf{N}-\mathsf{S}+1)}}. \tag{29}$$

Notice that $1 > \frac{\mathsf{M}}{\mathsf{N}} > \frac{\mathsf{M}-1}{\mathsf{N}-1} > \ldots > \frac{\mathsf{M}-\mathsf{S}+1}{\mathsf{N}-\mathsf{S}+1}$, it can be verified that β attains its maximum when $\mathsf{S} = 2$. Under the feasibility condition $\mathsf{S} \geq \mathsf{N} - \mathsf{N_r} + 2$, this implies $\mathsf{N} = \mathsf{N_r}$. Substituting these conditions into the ratio yields:

$$\frac{R}{R^*} \leq \frac{\mathsf{N} + \mathsf{M}}{\mathsf{N}} \leq 2, \tag{30}$$

which completes the proof.

Appendix B
Encodability Proof

In this appendix, we prove that the proposed construction guarantees encodability, i.e., for each dataset D_i, the submatrix $\mathbf{C}(\overline{\mathcal{D}_i}, \mathcal{O})$ has full row rank.

Recall that each uncoded groupwise key is partitioned into α independent pieces, and to match the key pieces, the key related part of the demand matrix is chosen as $\mathbf{F}_3 = \mathbf{I}_{\alpha\binom{\mathsf{N}}{\mathsf{S}}}$. Thus, each key piece corresponds to a distinct orthogonal column vector.

Consider an arbitrary set of servers $\mathcal{N} \subseteq [\mathsf{N}]$. Let $\Omega_{\mathcal{N}}$ denote the number of keys available to servers in $\mathcal{N}$, which means $\Omega_{\mathcal{N}} = |\mathcal{S}_{\mathcal{N}}|$ and $\mathcal{S}_{\mathcal{N}} = \cup_{n \in \mathcal{N}} \mathcal{S}_n$. We have the following lemma.

Lemma 2 (Available independent vectors): The number of available independent vectors to $\mathbf{C}(\mathcal{N}, \mathcal{O})$ is $\alpha\Omega_{\mathcal{N}}$.

Proof. For servers in $\mathcal{N}$, $\overline{\mathcal{S}_{\mathcal{N}}} = \cap_{n \in \mathcal{N}} \overline{\mathcal{S}_n}$ contains the keys unavailable to all servers in $\mathcal{N}$, and let $\overline{\mathcal{H}_{\mathcal{N}}}$ index the columns of $\mathbf{F}_3$ corresponding to these key pieces. Thus, the encodability constraint requires

$$\mathbf{C}(\mathcal{N}, \mathcal{O})\mathbf{F}_3(\mathcal{O}, \overline{\mathcal{H}_{\mathcal{N}}}) = \mathbf{0}. \tag{31}$$

From the above equation, we notice that the rows of $\mathbf{C}(\mathcal{N}, \mathcal{O})$ must lie in the left nullspace of $\mathbf{F}_3(\mathcal{O}, \overline{\mathcal{H}_{\mathcal{N}}})$. Since $\mathbf{F}_3$ is an identity matrix, its columns form an orthogonal basis. Then the left nullspace of $\mathbf{F}_3(\mathcal{O}, \overline{\mathcal{H}_{\mathcal{N}}})$ is formed by the column basis vectors in $\mathbf{F}_3(\mathcal{O}, \mathcal{H}_{\mathcal{N}})$, and thses column basis vectors are corresponding to keys in the set $\mathcal{S}_{\mathcal{N}}$. Each key is partitioned into α pieces, corresponding to α independent vectors. Hence, the number of available independent vectors is $\alpha\Omega_{\mathcal{N}}$. $\square$

We next compute $\Omega_{\mathcal{N}}$.

Lemma 3 (Available key number): For a server set $\mathcal{N} \subseteq [\mathsf{N}]$,

$$\Omega_{\mathcal{N}} = \binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{N} - |\mathcal{N}|}{\mathsf{S}}, \tag{32}$$

and equivalently,

$$\Omega_{\mathcal{N}} = \sum_{k=1}^{\mathsf{S}} \binom{|\mathcal{N}|}{k} \binom{\mathsf{N} - |\mathcal{N}|}{\mathsf{S} - k}. \tag{33}$$

Proof of (32). We focus on the servers in the set $\mathcal{N}$. There are $\binom{\mathsf{N}}{\mathsf{S}}$ uncoded groupwise keys in total. A key is unavailable to $\mathcal{N}$ if and only if it is shared exclusively among the remaining $\mathsf{N} - |\mathcal{N}|$ servers. The number of such keys is therefore $\binom{\mathsf{N}-|\mathcal{N}|}{\mathsf{S}}$. Subtracting these unavailable keys from the total yields (32).

Proof of (33). Each key is shared by exactly S servers. We classify keys according to the number of servers they are assigned to within the set $\mathcal{N}$. For a given $k \in \{1, \ldots, \mathsf{S}\}$, the number of keys shared by exactly k servers in $\mathcal{N}$ and $\mathsf{S} - k$ servers in $[\mathsf{N}] \setminus \mathcal{N}$ is $\binom{|\mathcal{N}|}{k}\binom{\mathsf{N}-|\mathcal{N}|}{\mathsf{S}-k}$. Summing over all possible values of k gives (33). $\square$

We now specialize to the server set associated with dataset D_i. In our model, each dataset is replicated at least M times, and therefore is not replicated at at most $\mathsf{N} - \mathsf{M}$ servers, meaning $|\overline{\mathcal{D}_i}| \leq \mathsf{N} - \mathsf{M}$. We first consider the case $|\overline{\mathcal{D}_i}| = \mathsf{N} - \mathsf{M}$; later we will show that the result also holds for all cases with $|\overline{\mathcal{D}_i}| < \mathsf{N} - \mathsf{M}$.

The matrix $\mathbf{C}(\overline{\mathcal{D}_i}, \mathcal{O})$ has dimension

$$(\mathsf{N} - \mathsf{M})r \times \alpha\binom{N}{S}, \tag{34}$$

where each of the $\mathsf{N} - \mathsf{M}$ servers contributes r rows, and $r = \left(\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{M}}{\mathsf{S}}\right)$ from Theorem 1. In our general scheme, each key is partitioned into $\mathsf{N} - \mathsf{M}$ pieces. From Lemma 2 and Lemma 3, the number of available independent vectors for $\mathbf{C}(\overline{\mathcal{D}_i}, \mathcal{O})$ is

$$\alpha\Omega_{\overline{\mathcal{D}_i}} = (\mathsf{N} - \mathsf{M})\left(\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{M}}{\mathsf{S}}\right). \tag{35}$$

Hence, the number of available independent vectors exactly matches the required row dimension of $\mathbf{C}(\overline{\mathcal{D}_i}, \mathcal{O})$.

Finally, since each server generates rows by random linear combinations of accessible vectors, the rows of $\mathbf{C}(\overline{\mathcal{D}_i}, \mathcal{O})$ are linearly independent with high probability, completing the encodability proof. $\square$

We now consider the case $|\overline{\mathcal{D}_i}| = x < \mathsf{N} - \mathsf{M}$. In this case, the matrix $\mathbf{C}(\overline{\mathcal{D}_i}, \mathcal{O})$ has dimension $xr \times \alpha\binom{N}{S}$, while the number of available independent vectors is $(\mathsf{N} - \mathsf{M})\left(\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{N}-x}{\mathsf{S}}\right)$. It therefore suffices to show that

$$(\mathsf{N} - \mathsf{M})\left(\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{N} - x}{\mathsf{S}}\right) \geq xr. \tag{36}$$

Equivalently, it is enough to show that the expression $\frac{\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{N}-x}{\mathsf{S}}}{x}$ is strictly decreasing in x, since $\frac{\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{N}-x}{\mathsf{S}}}{x} = \frac{r}{\mathsf{N}-\mathsf{M}}$

when $x = \mathsf{N} - \mathsf{M}$. This holds if

$$\frac{\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{N}-x}{\mathsf{S}}}{x} > \frac{\binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{N}-(x+1)}{\mathsf{S}}}{x+1}. \tag{37}$$

Rearranging terms yields

$$\binom{\mathsf{N}}{\mathsf{S}} > (x+1)\binom{\mathsf{N}-x}{\mathsf{S}} - x\binom{\mathsf{N}-x-1}{\mathsf{S}}. \tag{38}$$

Using the binomial identity $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$, the right side can be rewritten as

$$\binom{\mathsf{N}-x-1}{\mathsf{S}} + (x+1)\binom{\mathsf{N}-x-1}{\mathsf{S}-1}. \tag{39}$$

To verify the inequality, observe that

$$\begin{aligned}\binom{\mathsf{N}}{\mathsf{S}} &\overset{(33)}{=} \binom{\mathsf{N}-x-1}{\mathsf{S}} + \sum_{k=1}^{\mathsf{S}} \binom{x+1}{k}\binom{\mathsf{N}-x-1}{\mathsf{S}-k} \\ &> \binom{\mathsf{N}-x-1}{\mathsf{S}} + (x+1)\binom{\mathsf{N}-x-1}{\mathsf{S}-1},\end{aligned} \tag{40}$$

where the strict inequality follows since the summation contains additional positive terms for $k \geq 2$. This establishes the desired monotonicity and completes the argument for all $|\overline{\mathcal{D}_i}| < \mathsf{N} - \mathsf{M}$.

Appendix C
Decodability Proof

We prove that the proposed scheme satisfies the decodability constraint. Each server transmits r linear combinations, and hence the matrix $\mathbf{C}_{\mathcal{U}}$ has dimension $r\mathsf{N_r} \times r\mathsf{N_r}$, where $r = \binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{M}}{\mathsf{S}}$. By construction, the columns of $\mathbf{C}_{\mathcal{U}}$ consist of two types: the first n columns correspond to gradient-related components, while the remaining $(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}$ columns correspond to key-related components. The gradient-related columns are generated independently at random and are therefore full rank with high probability. Consequently, it suffices to show that the key-related columns are linearly independent.

Let $\mathbf{C}(\mathcal{U}, \mathcal{O})$ denote the submatrix of $\mathbf{C}_{\mathcal{U}}$ formed by these key-related columns. Since each groupwise key is partitioned into $\mathsf{N}-\mathsf{M}$ independent pieces and the key-related demand matrix is chosen as an identity matrix, the encodability constraints imply that the rows of $\mathbf{C}(\mathcal{U}, \mathcal{O})$ must lie in the subspace spanned by the basis vectors corresponding to keys available to servers in $\mathcal{U}$. As shown in Appendix B, each such key contributes exactly $\mathsf{N} - \mathsf{M}$ independent dimensions. Let $\Omega_{\mathcal{U}}$ denote the number of keys available to the servers in $\mathcal{U}$; then the number of available independent vectors for $\mathbf{C}(\mathcal{U}, \mathcal{O})$ is given by $(\mathsf{N} - \mathsf{M})\Omega_{\mathcal{U}}$.

By Lemma 3 in Appendix B, the number of keys available to the servers in $\mathcal{U}$ is

$$\Omega_{\mathcal{U}} = \binom{\mathsf{N}}{\mathsf{S}} - \binom{\mathsf{N}-\mathsf{N_r}}{\mathsf{S}}. \tag{41}$$

Under the feasibility condition $\mathsf{S} \geq \mathsf{N} - \mathsf{N_r} + 2$, we have $\binom{\mathsf{N}-\mathsf{N_r}}{\mathsf{S}} = 0$, and therefore $\Omega_{\mathcal{U}} = \binom{N}{S}$. It follows that the number of available independent vectors for $\mathbf{C}(\mathcal{U}, \mathcal{O})$ is exactly $(\mathsf{N} - \mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}$, which matches the number of columns of $\mathbf{C}(\mathcal{U}, \mathcal{O})$.

Since all servers in $\mathcal{U}$ possess all available keys, the corresponding key-induced subspace can be assigned to the rows of $\mathbf{C}(\mathcal{U}, \mathcal{O})$ without overlap, ensuring that these columns are linearly independent. Combined with the full-rank property of the gradient-related columns, this implies that $\mathbf{C}_{\mathcal{U}}$ is invertible.

Hence, the user can decode from the transmissions of any N_r servers, completing the proof.

Appendix D
Security Proof

We first present the security proof for the illustrative example, where the objective is to verify that $I(X_1, X_2, X_3; g_1, g_2, g_3 | g_1 + g_2 + g_3) = 0$. We partition the matrix $\mathbf{W}$ into two submatrices $\mathbf{W} = [\mathbf{W}_G, \mathbf{W}_Q]^T$, where $\mathbf{W}_G$ consists of the first $n\mathsf{K}$ rows of $\mathbf{W}$ corresponding to the gradient segments, and $\mathbf{W}_Q$ comprises the remaining $(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}$ rows corresponding to the key segments. Assuming $\mathsf{L} = 1$, we proceed as follows:

$$\begin{aligned}
&I(X_1, X_2, X_3; g_1, g_2, g_3 | g_1 + g_2 + g_3) && \text{(42a)}\\
=&H(X_1, X_2, X_3 | g_1 + g_2 + g_3) - H(X_1, X_2, X_3 | g_1, g_2, g_3) && \text{(42b)}\\
=&H(X_1, X_2, X_3 | g_1 + g_2 + g_3) - H(\mathbf{F}_3\mathbf{W}_Q | g_1, g_2, g_3) && \text{(42c)}\\
=&H(X_1, X_2, X_3 | g_1 + g_2 + g_3) - H(\mathbf{F}_3\mathbf{W}_Q) && \text{(42d)}\\
=&H(X_1, X_2, X_3 | g_1 + g_2 + g_3) - 1 && \text{(42e)}\\
=&H(X_1, X_2, X_3, g_1 + g_2 + g_3) - H(g_1 + g_2 + g_3) - 1 && \text{(42f)}\\
=&H(X_1, X_2, X_3) - H(g_1 + g_2 + g_3) - 1 && \text{(42g)}\\
=&H(X_1, X_2, X_3) - 2 && \text{(42h)}\\
\leq&H(X_1) + H(X_2) + H(X_3) - 2 && \text{(42i)}\\
\leq&\frac{2}{3} \times 3 - 2 = 0, && \text{(42j)}
\end{aligned}$$

where (42d) follows from the mutual independence of the gradients and the keys; (42e) holds because $\mathbf{F}_3\mathbf{W}_Q$ contains three independent key symbols, each of length $\frac{\mathsf{L}}{3}$; (42g) relies on the fact that the transmitted messages (X_1, X_2, X_3) are sufficient to recover the sum $g_1+g_2+g_3$; and (42j) accounts for the total transmission size, as each server transmits r messages, each of length $\frac{\mathsf{L}}{3}$.

Next, we provide the security proof for the general case, where we must establish $I\left(X_1, \ldots, X_\mathsf{N}; g_1, \ldots, g_\mathsf{K} | \sum_{k\in[\mathsf{K}]} g_k\right) = 0$. Analogous to the example, we obtain:

$$\begin{aligned}
&I\left(X_1, \ldots, X_\mathsf{N}; g_1, \ldots, g_\mathsf{K} | \sum_{k\in[\mathsf{K}]} g_k\right) && \text{(43a)}\\
=&H\left(X_1, \ldots, X_\mathsf{N} | \sum_{k\in[\mathsf{K}]} g_k\right) - H\left(X_1, \ldots, X_\mathsf{N} | g_1, \ldots, g_\mathsf{K}\right) && \text{(43b)}\\
=&H\left(X_1, \ldots, X_\mathsf{N} | \sum_{k\in[\mathsf{K}]} g_k\right) - H\left(\mathbf{F}_3\mathbf{W}_Q | g_1, \ldots, g_\mathsf{K}\right) && \text{(43c)}
\end{aligned}$$

$$=H\left(X_1,\ldots,X_{\mathsf{N}}|\sum_{k\in[\mathsf{K}]}g_k\right)-H\left(\mathbf{F}_3\mathbf{W}_Q\right) \tag{43d}$$

$$=H\left(X_1,\ldots,X_{\mathsf{N}}|\sum_{k\in[\mathsf{K}]}g_k\right)-\frac{(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}}{n} \tag{43e}$$

$$\leq H\left(X_1,\ldots,X_{\mathsf{N}},\sum_{k\in[\mathsf{K}]}g_k\right)-H\left(\sum_{k\in[\mathsf{K}]}g_k\right)-\frac{(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}}{n} \tag{43f}$$

$$=H\left(X_{\mathcal{U}(1)},\ldots,X_{\mathcal{U}(\mathsf{N_r})},\sum_{k\in[\mathsf{K}]}g_k\right)-1-\frac{(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}}{n} \tag{43g}$$

$$=H\left(X_{\mathcal{U}(1)},\ldots,X_{\mathcal{U}(\mathsf{N_r})}\right)-1-\frac{(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}}{n} \tag{43h}$$

$$\leq\sum_{i\in[\mathcal{U}]}H\left(X_i\right)-1-\frac{(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}}{n} \tag{43i}$$

$$\leq\frac{r}{n}\mathsf{N_r}-1-\frac{(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}}{n}=0, \tag{43j}$$

where (43d) results from the independence of the gradients and the keys; (43e) follows because $\mathbf{F}_3\mathbf{W}_Q$ comprises $(\mathsf{N}-\mathsf{M})\binom{\mathsf{N}}{\mathsf{S}}$ key pieces, each of length $\frac{\mathsf{L}}{n}$; (43g) is justified by the decodability property of $\mathbf{C}$, which ensures that transmissions from any set of servers $\mathcal{U}\in\binom{\mathsf{N}}{\mathsf{N_r}}$ contain all necessary information; (43h) holds because the sum of gradients $\sum_{k\in[\mathsf{K}]}g_k$ is recoverable from $(X_{\mathcal{U}(1)},\ldots,X_{\mathcal{U}(\mathsf{N_r})})$; and (43j) reflects that each server transmits r messages, each of length $\frac{\mathsf{L}}{n}$. This concludes the proof of security for the proposed scheme.

Appendix E
Derivation of Communication Cost

We derive the communication cost by identifying the conditions under which the linear system induced by the proposed scheme is solvable.

Assume a communication cost of $\mathsf{R}=\frac{r}{n}$, where each server transmits r independent messages and each gradient is partitioned into n segments. Each key is further partitioned into α segments. Under this construction, the coding matrix $\mathbf{C}$ has dimension $r\mathsf{N}\times r\mathsf{N_r}$, and the demand matrix $\mathbf{F}$ has dimension $(n+\alpha\binom{\mathsf{N}}{\mathsf{S}})\times(n\mathsf{K}+\alpha\binom{\mathsf{N}}{\mathsf{S}})$. For the product $\mathbf{CF}$ to be well defined, the number of columns of $\mathbf{C}$ must equal the number of rows of $\mathbf{F}$, yielding the constraint

$$r\mathsf{N_r}=n+\alpha\binom{\mathsf{N}}{\mathsf{S}}. \tag{44}$$

Next, consider the submatrix $\mathbf{C}(\overline{\mathcal{D}_i},\mathcal{O})$, which has dimension $(\mathsf{N}-\mathsf{M})r\times\alpha\binom{\mathsf{N}}{\mathsf{S}}$. By Lemma 2, the dimension of the available subspace for this matrix is $\alpha\Omega_{\overline{\mathcal{D}_i}}$, and by Lemma 3 $\Omega_{\overline{\mathcal{D}_i}}=\binom{\mathsf{N}}{\mathsf{S}}-\binom{\mathsf{M}}{\mathsf{S}}$. For the corresponding linear system to be solvable, the available subspace dimension must be at least as large as the number of columns, leading to the necessary condition

$$\alpha\left(\binom{\mathsf{N}}{\mathsf{S}}-\binom{\mathsf{M}}{\mathsf{S}}\right)\geq(\mathsf{N}-\mathsf{M})r. \tag{45}$$

Finally, eliminating α using (44) and (45), we obtain the following lower bound on the communication cost:

$$\mathsf{R}=\frac{r}{n}\geq\frac{\binom{\mathsf{N}}{\mathsf{S}}-\binom{\mathsf{M}}{\mathsf{S}}}{\left(\binom{\mathsf{N}}{\mathsf{S}}-\binom{\mathsf{M}}{\mathsf{S}}\right)\mathsf{N_r}-\binom{\mathsf{N}}{\mathsf{S}}(\mathsf{N}-\mathsf{M})}. \tag{46}$$